BUSINESS REQUIREMENT DOCUMENT

NAME: MAHWISH IFTKHAR

STD CODE: B2255R10106231

PROJECT: QR BASED VISITOR MANAGEMENT SYSTEM FOR OFFICES

COURSE: B.TECH (CSE)

SEM: 8th

1.Introduction : The QR-Based Visitor Management System is designed to digitize and automate visitor entry processes using QR codes. It replaces manual registers with a secure, efficient, and contactless solution.

1.1 Business Objectives

- Improve security and visitor tracking
- Reduce manual paperwork
- Enable contactless check-in
- Maintain digital records for audits

1.2 Stakeholders

- Visitors
- Security Guards
- Admin/Management

- IT Team

1.3 Scope

In Scope

- Visitor registration
- QR code generation
- Entry/Exit tracking
- Admin dashboard

1.4 Out of Scope

- Biometric systems
- Payment features
- Advanced AI analytics

1.5 Business Requirements

- System must allow visitor pre-registration
- System must generate unique QR codes
- System must track entry and exit time
- Admin must view and manage records

1.6 Success Criteria

- Reduced check-in time
- Improved data accuracy

- Secure visitor records

2. Software Requirements Specification (SRS)

2.1 Introduction

2.1.1 Purpose

This document defines the functional and non-functional requirements of the QR-Based Visitor Management System.

2.1.2 Scope

The system enables visitor registration, QR generation, scanning, and tracking via a web-based interface.

2.2 Overall Description

2.2.1 Product Perspective

- Web-based system
- Works with browser + camera
- Backend handles QR and database

2.2.2 User Classes

- Visitor → registers and gets QR
- Guard → scans QR
- Admin → monitors system

2.3 System Features

2.3.1 Visitor Registration

- Input: Name, phone, purpose
- Output: QR Code

2.3.2 QR Code Generation

- Unique QR per visitor
- Stored in database

2.3.3 QR Scanning

- Scan via camera
- Validate visitor
- Record entry/exit

2.3.4 Admin Dashboard

- View visitor logs
- Filter/search records

3. Functional Requirements

FR1: Visitor Registration

- System shall allow visitors to enter details
- System shall validate input fields

FR2: QR Code Generation

- System shall generate a unique QR code
- QR must be linked to visitor record

FR3: Entry Logging

- System shall record check-in time on scan

FR4: Exit Logging

- System shall record check-out time

FR5: Admin Access

- System shall provide login for admin
- Admin shall view visitor data

FR6: Data Storage

- System shall store all records in database

4. Non-Functional Requirements

4.1 Performance

- QR scan response time < 2 seconds
- System should handle multiple users

4.2 Security

- Admin authentication required
- Secure API communication (HTTPS)
- Prevent duplicate QR misuse

4.3 Usability

- Simple interface for guards
- Mobile-friendly design

4.4 Reliability

- System uptime ≥ 99%
- Data backup support

4.5 Scalability

- Should support increasing visitors
- Easily upgradeable

4.6 Maintainability

- Modular code structure
- Easy bug fixing and updates

5. System Architecture

- Frontend: React.js
- Backend: Fast API
- Database: SQLite / MySQL
- QR Generation: Python library
- Scanner: Camera/Web

6. Data Requirements

Field	Description
Name	Visitor Name
Phone	Contact Number
Purpose	Reason for visit
QR ID	Unique identifier

Check-in	Entry time
Check-out	Exit time

7. Process Flow

1. Visitor registers
2. System generates QR
3. Visitor arrives
4. Guard scans QR
5. Entry logged
6. Exit scan logged
7. Data stored

8. Assumptions & Constraints

Assumptions

- Users have smartphones
- Internet available

Constraints

- Depends on camera quality
- Network dependency

9. Future Enhancements

- Face recognition

- SMS/Email alerts
- Mobile app version
- Multi-location support